

Recommended Tech Stack

AI-Powered Portfolio Web App with Personal Sidekick

This stack is optimized for **performance, scalability, modern UX, and AI integration** while staying maintainable for a solo developer.

1. Frontend

Framework

Next.js (App Router)

- Server components for performance
- API routes for secure AI calls
- SEO-friendly for recruiter discovery
- Easy Vercel deployment

Styling

Tailwind CSS

- Fast UI development
- Easy responsive design
- Works well with design systems

UI Components

shadcn/ui

- Accessible, modern components
- Easy to customize to match Shopify-style visuals

Animation

Framer Motion

- Smooth entrance animations
- Scroll-triggered transitions
- Interactive hover effects

Optional advanced visuals: - **Three.js** or **Spline** for hero background visuals

2. AI Sidekick Layer

LLM Integration

OpenAI API (GPT-4.1 / GPT-4o class models)

- High-quality conversational responses
- Strong instruction-following for personality shaping

Context Injection

Custom Knowledge Layer (RAG-lite approach) - Store resume, bio, and portfolio data as structured JSON or markdown - Inject relevant context into prompts per query

Server Handling

Next.js Server Actions or API Routes

- Keep API keys secure
 - Handle prompt construction server-side
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3. Content & Data Management

Portfolio Content

Local Markdown or JSON Files

- Easy to update without database complexity
- Perfect for projects, experience, bio

OR (if you want CMS): **Sanity / Contentful / Notion API**

Resume Storage

- Structured JSON extracted from resume
 - Also allow PDF download
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4. GitHub Integration

GitHub REST or GraphQL API

- Fetch repositories dynamically
- Filter “featured” projects
- Display stars, tech, descriptions

Cache results using: - Next.js caching or ISR

5. Backend / Server

You may not need a traditional backend.

Use **Next.js as a full-stack framework**: - API routes for AI - Data fetching for GitHub - Form handling (contact form)

If needed later: **Node.js + Express** or **Supabase Edge Functions**

6. Database (Optional)

If storing analytics or chat logs:

Supabase (Postgres)

- Easy auth if needed later
- Simple DB + storage
- Serverless friendly

OR lightweight option: **Firebase**

7. Authentication (Optional Future)

If you add recruiter dashboards or admin panels:

- **Clerk** or **Auth.js (NextAuth)**
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8. Deployment & DevOps

Hosting

Vercel

- Native Next.js support
- Edge functions
- Easy environment variable management

Monitoring

- Vercel Analytics
 - Sentry for error tracking
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9. Performance Optimization

- Next.js Image optimization
 - Lazy load animations
 - Use server components where possible
 - Cache GitHub API responses
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10. Summary Stack

Layer	Technology
Frontend	Next.js + Tailwind + shadcn/ui
Animation	Framer Motion
AI	OpenAI API
Data	JSON/Markdown or Sanity
GitHub	GitHub API
Backend	Next.js API routes
Database (optional)	Supabase
Hosting	Vercel

This stack keeps the project **modern, scalable, AI-ready, and visually premium** while staying realistic for solo development.

End of Tech Stack Recommendation